

Alejandro Zumbado Vargas

Full Stack Developer

Poás, Alajuela, Costa Rica | +506 6022-1640 | aazv.ale@gmail.com |
github.com/alejandroZumbado | linkedin.com/in/alejandro-zumbado-4124661ab

SUMMARY

Computer Engineer (TEC, 2019) with 4+ years building end-to-end web products: React/Next.js frontends, Django/Node.js backends, REST APIs, and LLM integrations. Currently at Opal Interactive, building projects for various clients (XR simulators, kiosk/AR apps, AI avatars). In parallel, I design and ship my own products — an AI-driven autonomous content pipeline (newspulse.dev) and published multiplayer WebGL games — owning design, development, deployment (CI/CD), and maintenance.

SKILLS

Languages
JavaScript · TypeScript · Python · C#

Frontend
React · Next.js · Astro · Three.js / WebGL · HTML · CSS

Backend
Node.js · Express · NestJS · Django · REST APIs

AI / Data
LLM integration (Gemini) · RAG pipelines · content automation

DevOps
Git/GitHub · GitHub Actions (CI/CD) · Vercel · Firebase · pnpm workspaces

WORK EXPERIENCE

Full-Stack Developer — Opal Interactive
Apr 2026 — Present

Studio that takes on projects from various companies and distributes them across its dev team

- Opal Web Avatar: interactive AI avatar platform — React 19, Express, lipsync, RAG.
- HealthSim: Unity XR training simulator for medical equipment, with remote sessions via Netcode + Unity Relay.
- totempanadol: Unity kiosk/AR app with thermal camera, face detection, and Firebase.
- Contributed to Startus: a 4-repo SaaS platform (NestJS, React, FastAPI, Chrome extension).

React / Frontend Developer — Freelance — independent projects
2021 — Present

Multiple clients and projects

- Solucom — Remote Gate Access (2023): IoT access-control system with a Raspberry Pi (Python) as the physical controller, Vue.js frontend, and Laravel/PHP backend, synced over an API with a password database.
- Built applications and components with React: clean architecture, reusable components, responsive design.
- Integrated third-party APIs, fixed bugs, and improved performance under tight deadlines.

Full-Stack Developer (Jr/Mid) — IMMERSO
Apr 2021 — Jan 2023

- Built and maintained frontends with React.js and Next.js connected to Django backends.
- Built interactive web and VR experiences with A-Frame.js: editors, maps, and database-driven content.

Full-Stack Developer (Jr/Mid) — WOW Emotions Jan 2022 — Feb 2023

- Led end-to-end development of a VR-enabled website for the 'Doit' reconstruction project.
- Designed and built 'Giz', an immersive nature experience with 5+ VR levels.
- Optimized performance and visual consistency across devices and browsers.

Full-Stack Developer (Jr) — SolvoSoft Aug 2020 — Feb 2021

- Developed and optimized web features with JavaScript and Django on legacy systems.
- Collaborated with cross-functional teams to improve UI performance and responsiveness.

INDEPENDENT PROJECTS

NewsPulse — personal product — newspulse.dev Jun 2026 — Present

Fully autonomous, AI-generated news pipeline

- Python pipeline that researches news via RSS, writes articles in 4 languages with an LLM, and fact-checks them against source material before publishing.
- Static frontend built with Astro + Tailwind, deployed on Vercel; publishing runs automatically on a GitHub Actions cron with no human in the loop.

Tinted Showdown — personal game — published 2025 — Present

2–4 player multiplayer, PC / Android / WebGL

- Multiplayer Unity game with Netcode for GameObjects + Unity Relay; live, playable WebGL build.
- Extracting the networking layer as a standalone package (QuickLobby — Netcode Relay Kit), in development for release as a Unity asset.

Cannons — personal game — published Aug 2025 — Present

Tower-defense, 500 levels

- Unity tower-defense with progression across up to 500 levels and a self-play Python AI that plays and designs new levels.

FeedBack Figuras 3D — personal product 2025 — Present

Customer feedback collection system

- Next.js 15 + React 19 app for rating custom 3D-printed figures by category, uploading photos, and exporting results from an admin panel.
- Backend with Supabase, validation with Zod and React Hook Form, animations with Framer Motion.

EDUCATION

Computer Engineering 2019

Tecnológico de Costa Rica (TEC)

High School Diploma 2015

Liceo San Rafael de Alajuela

LANGUAGES

Spanish — Native

English — Intermediate